Towards Scalable and Reliable Capsule Networks for Challenging NLP Applications

Wei Zhao[1]; Haiyun Peng[2]; Steffen Eger[1]; Eric Cambria[2]; Min Yang[3]

[1]Technische Universitat Darmstadt, [2]Nanyang Technological University, [3]Chinese Academy of Science

AIPHES

www.aiphes.tu-darmstadt.de

TECHNISCHE UNIVERSITÄT DARMSTADT

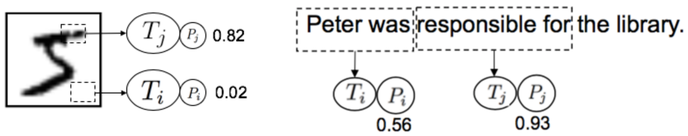
Summary

+ Capsules Network promises better generalization ability of neural models for NLP applications.
- The scalability and reliability of Capsule Network remain a challenge

Our contributions:

- Introduce an agreement score to evaluate the performance of routing process at instance level
- Present an adaptive optimizer to strengthen the reliability of routing
- Introduce capsule compression and partial routing to improve scalability

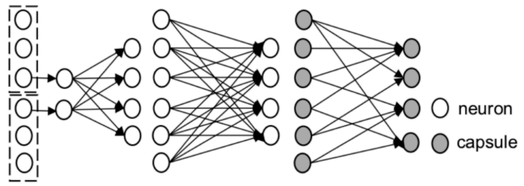
Capsule Network



T_i : holds one vector for the information of an object or words

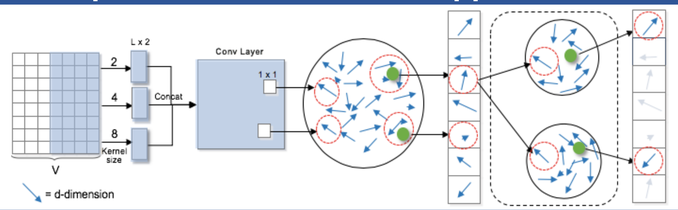
P_i : shows if it is useful for the task at hand

Information aggregation between layers:

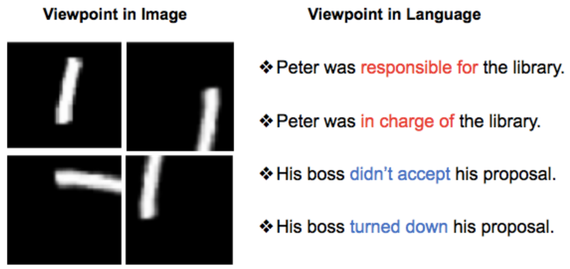


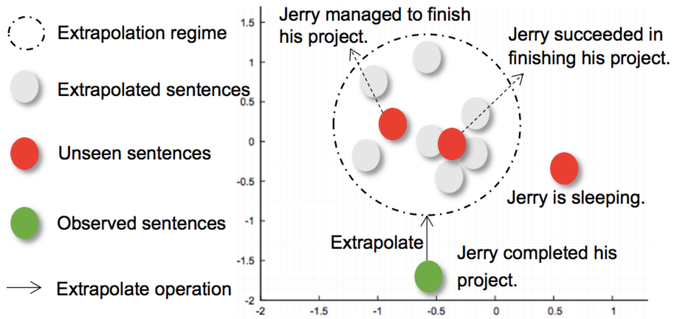
a) Pooling Connection b) Full Connection c) Routed Connection

Capsule Network for NLP Applications



Generalization and Extrapolation





Instance Level Routing

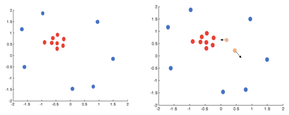
Routing Connection between layers:

$$\{\mathbf{u}_{ij} \in \mathbb{R}^d\}_{i=1,\dots,H,j=1\dots,N} \mapsto \{\mathbf{v}_j \in \mathbb{R}^d\}_{j=1}^N$$

Negative agreement score: $\min_{c,v} f(u) = -\sum_{i,j} c_{ij}\langle v_j, u_{j|i}\rangle$

Kernel agreement score: $\min_{c,v} f(u) = -\sum_{i,j} c_{ij} k(d(v_j, u_{j|i}))$

Partial routing: $\min_{c,v} -\sum_i \Big(\sum_{j\in D^+} c_{ij}\langle v_j, u_{j|i}\rangle + \sum_{k\in D^-} c_{ik}\langle v_k, u_{k|i}\rangle\Big)$



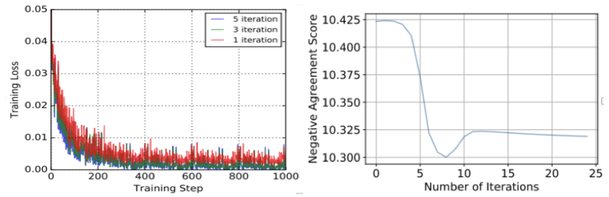
Blue dots: draws from uniform distribution

Red dots: lie in Gaussian distribution

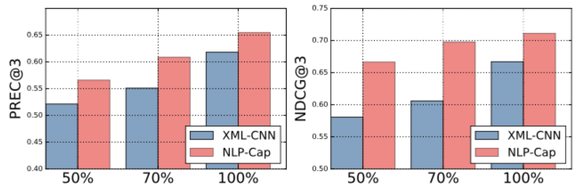
Orange dots: need more iterations

Experiments on MLC and QA

Compare routing process at system and instance level

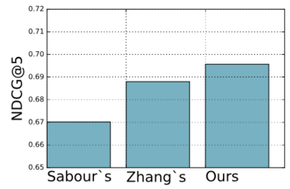


Generalize from different fractions of training examples



Performance on different routing processes

Method	PREC@1	PREC@3	PREC@5
XML-CNN	75.65	61.81	50.90
NLP-Capsule + Sabour's Routing	79.14	64.33	51.85
NLP-Capsule + Zhang's Routing	80.20	65.48	52.83
NLP-Capsule + Our Routing	**80.62**	**65.61**	**53.66**
Method	**NDCG@1**	**NDCG@3**	**NDCG@5**
XML-CNN	75.65	66.71	64.45
NLP-Capsule + Sabour's Routing	79.14	70.13	67.02
NLP-Capsule + Zhang's Routing	80.20	71.11	68.80
NLP-Capsule + Our Routing	**80.62**	**71.34**	**69.57**



Closing thoughts

Takeaways:

- Capsule Networks are a powerful class of neural networks with good generalization ability
- Routing at Instance level has performance gains
- Try out our Capsule Network with better scalability and reliability

https://github.com/AIPHES/acl19-generalization-capsule

Open problems:

- Introspect Capsule Network for reasons of performance gains
- Can we apply Capsule Network to language modeling and text generation

References

1. Wei Zhao, Haiyun Peng, Steffen Eger, Erik Cambria, Min Yang. Towards Scalable and Reliable Capsule Networks for Challenging NLP Applications, **ACL**, 2019
2. Wei Zhao, Jianbo Ye, Min Yang, Zeyang Lei, Soufei Zhang, Zhou Zhao, Investigating Capsule Networks with Dynamic Routing for text classification, **EMNLP**, 2018
3. Soufei Zhang, Wei Zhao, Xiaofu Wu, Quan Zhou, Fast Dynamic Routing Based on Weighted Kernel Density Estimation, **Concurrency and Computation: Practice and Experience**, 2019

NLLG

NANYANG TECHNOLOGICAL UNIVERSITY

CHINESE ACADEMY OF SCIENCES

© POSTER TEMPLATE BY GENIGRAPHICS® 1.800.790.4001 WWW.GENIGRAPHICS.COM